

EXP 7 COMPUTERNETWORK

AIM: To implement a sliding window flow control protocol in C, where the sender transmits frames within a window through Sender_Buffer.txt and the receiver replies with ACK or NACK through Receiver_Buffer.txt.

Sender.c:

```
#include <stdio.h>

#include <stdlib.h>

#include <string.h>


#ifdef _WIN32
#include <windows.h>

void delay(int seconds)
{
    Sleep(seconds * 1000);
}
#else
#include <unistd.h>

void delay(int seconds)
{
    sleep(seconds);
}
#endif


#define SENDER_FILE "Sender_Buffer.txt"
#define RECEIVER_FILE "Receiver_Buffer.txt"


void clearFile(const char *filename)
{
    FILE *fp = fopen(filename, "w");

    if (fp != NULL)
        fclose(fp);
}
```

```
}
```

```
int readAcknowledgement(int *type, int *number)
```

```
{
```

```
    FILE *fp;
```

```
    char line[100];
```

```
    fp = fopen(RECEIVER_FILE, "r");
```

```
    if (fp == NULL)
```

```
        return 0;
```

```
    if (fgets(line, sizeof(line), fp) == NULL)
```

```
    {
```

```
        fclose(fp);
```

```
        return 0;
```

```
    }
```

```
    fclose(fp);
```

```
    if (sscanf(line, "ACK:%d", number) == 1)
```

```
    {
```

```
        *type = 1;
```

```
        return 1;
```

```
    }
```

```
    if (sscanf(line, "NACK:%d", number) == 1)
```

```
    {
```

```
        *type = 2;
```

```
        return 1;
```

```
    }
```

```
    return 0;
}

void writeFrames(char *message, int start, int end)
{
    FILE *fp;

    fp = fopen(SENDER_FILE, "w");

    if (fp == NULL)
    {
        printf("Error: Cannot open Sender_Buffer.txt\n");
        exit(1);
    }

    printf("\n-----\n");
    printf("Sending frames %d to %d\n", start, end);
    printf("-----\n");

    for (int i = start; i <= end; i++)
    {
        fprintf(fp, "Frame:%d DATA:%c\n", i, message[i]);

        printf("Frame No: %d  DATA: %c\n",
            i,
            message[i]);
    }

    fclose(fp);
}
```

```
int main()
{
    char message[500];
    int windowSize;

    int totalFrames;
    int base = 0;

    int ackType;
    int ackNumber;

    printf("===== SLIDING WINDOW SENDER =====\n\n");

    printf("Enter window size: ");
    scanf("%d", &windowSize);

    if (windowSize <= 0)
    {
        printf("Invalid window size.\n");
        return 1;
    }

    getchar();

    printf("Enter text message: ");
    fgets(message, sizeof(message), stdin);

    message[strlen(message)] = '\0';

    totalFrames = strlen(message);
```

```
if (totalFrames == 0)
{
    printf("Message cannot be empty.\n");
    return 1;
}

clearFile(SENDER_FILE);
clearFile(RECEIVER_FILE);

printf("\nMessage: %s\n", message);
printf("Total frames: %d\n", totalFrames);
printf("Window size: %d\n", windowSize);

while (base < totalFrames)
{
    int end = base + windowSize - 1;

    if (end >= totalFrames)
        end = totalFrames - 1;

    writeFrames(message, base, end);

    printf("\nSender_Buffer.txt updated.");
    printf("\nWaiting for acknowledgement");

    while (1)
    {
        delay(2);

        if (readAcknowledgement(&ackType, &ackNumber))
```

```
{  
    if (ackType == 1)  
    {  
        printf("\nReceived ACK:%d\n", ackNumber);  
  
        if (ackNumber >= base)  
        {  
            base = ackNumber + 1;  
  
            clearFile(RECEIVER_FILE);  
  
            printf("Sliding window moved.\n");  
            break;  
        }  
    }  
    else if (ackType == 2)  
    {  
        printf("\nReceived NACK:%d\n", ackNumber);  
  
        printf("Resending frames from frame %d...\n",  
            ackNumber);  
  
        clearFile(RECEIVER_FILE);  
  
        if (ackNumber >= base)  
            base = ackNumber;  
  
        break;  
    }  
}
```

```
}

clearFile(SENDER_FILE);

printf("\n=====\\n");
printf("All frames transmitted successfully!\\n");
printf("Message transmission completed.\\n");
printf("=====\\n");

return 0;
}
```

Receiver.c:

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

#ifdef _WIN32
#include <windows.h>
void delay(int seconds)
{
    Sleep(seconds * 1000);
}
#else
#include <unistd.h>
void delay(int seconds)
{
    sleep(seconds);
}
```

```
#endif
```

```
#define SENDER_FILE "Sender_Buffer.txt"
```

```
#define RECEIVER_FILE "Receiver_Buffer.txt"
```

```
int readFrames(int *frameNumbers, char *data, int maxFrames)
```

```
{
```

```
    FILE *fp;
```

```
    char line[200];
```

```
    int count = 0;
```

```
    fp = fopen(SENDER_FILE, "r");
```

```
    if (fp == NULL)
```

```
        return 0;
```

```
    while (fgets(line, sizeof(line), fp) != NULL &&
```

```
        count < maxFrames)
```

```
    {
```

```
        int frameNo;
```

```
        char ch;
```

```
        if (sscanf(line, "Frame:%d DATA:%c",
```

```
            &frameNo, &ch) == 2)
```

```
        {
```

```
            frameNumbers[count] = frameNo;
```

```
            data[count] = ch;
```

```
            count++;
```

```
        }
```



```
}

fclose(fp);

return count;
}

void writeACK(int number)
{
    FILE *fp = fopen(RECEIVER_FILE, "w");

    if (fp == NULL)
        return;

    fprintf(fp, "ACK:%d\n", number);

    fclose(fp);
}

void writeNACK(int number)
{
    FILE *fp = fopen(RECEIVER_FILE, "w");

    if (fp == NULL)
        return;

    fprintf(fp, "NACK:%d\n", number);

    fclose(fp);
}
```

```
int main()
{
    int expectedFrame = 0;

    int frameNumbers[100];
    char data[100];

    int lastFirstFrame = -1;

    printf("===== SLIDING WINDOW RECEIVER =====\n\n");

    printf("Receiver is waiting for frames...\n");

    while (1)
    {
        int count = readFrames(frameNumbers, data, 100);

        if (count > 0)
        {
            /*
             * Process only a new Sender_Buffer window.
             */
            if (frameNumbers[0] != lastFirstFrame)
            {
                int valid = 1;

                printf("\n-----\n");
                printf("Frames received:\n");
                printf("-----\n");

                for (int i = 0; i < count; i++)
```

```
{  
    printf("Frame No: %d  DATA: %c\n",  
        frameNumbers[i],  
        data[i]);  
  
    if (frameNumbers[i] != expectedFrame)  
    {  
        printf("\nExpected Frame: %d\n",  
            expectedFrame);  
  
        printf("Received Frame: %d\n",  
            frameNumbers[i]);  
  
        printf("ERROR DETECTED!\n");  
  
        printf("Sending NACK:%d\n",  
            expectedFrame);  
  
        writeNACK(expectedFrame);  
  
        valid = 0;  
        break;  
    }  
  
    expectedFrame++;  
}  
  
if (valid)  
{  
    int ackNumber = expectedFrame - 1;
```

```
printf("\nAll frames received correctly.\n");

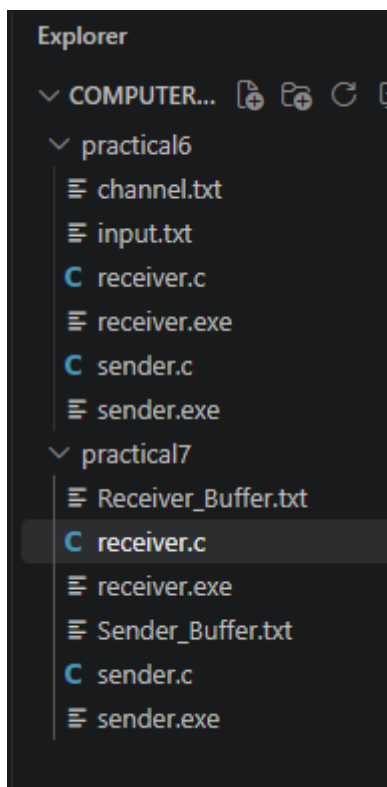
printf("Sending ACK:%d\n",
    ackNumber);

writeACK(ackNumber);
}

lastFirstFrame = frameNumbers[0];
}
}

delay(1);
}

return 0;
}
```



```
Problems  Output  Debug Console  Terminal  Ports

PS C:\Users\AI_LAB\Documents\COMPUTERNETWORK HAMMING CODE> cd "practical7"
PS C:\Users\AI_LAB\Documents\COMPUTERNETWORK HAMMING CODE\practical7> .\receiver.exe
===== SLIDING WINDOW RECEIVER =====

Receiver is waiting for frames...
█
```

```
PS C:\Users\AI_LAB\Documents\COMPUTERNETWORK HAMMING CODE> cd "practical7"
PS C:\Users\AI_LAB\Documents\COMPUTERNETWORK HAMMING CODE\practical7> gcc sender.c -o sender.exe
PS C:\Users\AI_LAB\Documents\COMPUTERNETWORK HAMMING CODE\practical7> .\sender.exe
===== SLIDING WINDOW SENDER =====

Enter window size: █
```

Sender:

```
PS C:\Users\AI_LAB\Documents\COMPUTERNETWORK HAMMING CODE> cd "
PS C:\Users\AI_LAB\Documents\COMPUTERNETWORK HAMMING CODE\pract
PS C:\Users\AI_LAB\Documents\COMPUTERNETWORK HAMMING CODE\pract
===== SLIDING WINDOW SENDER =====

Enter window size: 3
Enter text message: computer

Message: computer
Total frames: 8
Window size: 3

-----
Sending frames 0 to 2
-----
Frame No: 0   DATA: c
Frame No: 1   DATA: o
Frame No: 2   DATA: m

Sender_Buffer.txt updated.
Waiting for acknowledgement
Received ACK:2
Sliding window moved.
```

```
-----
Sending frames 3 to 5
-----
Frame No: 3   DATA: p
Frame No: 4   DATA: u
Frame No: 5   DATA: t

Sender_Buffer.txt updated.
Waiting for acknowledgement
Received ACK:5
Sliding window moved.

-----
Sending frames 6 to 7
-----
Frame No: 6   DATA: e
Frame No: 7   DATA: r

Sender_Buffer.txt updated.
Waiting for acknowledgement
Received ACK:7
Sliding window moved.
```

```
Problems  Output  Debug Console  Terminal  Ports

Frame No: 5   DATA: t

Sender_Buffer.txt updated.
Waiting for acknowledgement
Received ACK:5
Sliding window moved.

-----
Sending frames 6 to 7
-----

Frame No: 6   DATA: e
Frame No: 7   DATA: r

Sender_Buffer.txt updated.
Waiting for acknowledgement
Received ACK:7
Sliding window moved.

=====
All frames transmitted successfully!
Message transmission completed.
=====
PS C:\Users\AI_LAB\Documents\COMPUTERNETWORK HAMMING CODE\practical7> |
```

Receiver:

```
PS C:\Users\AI_LAB\Documents\COMPUTERNETWORK HAMMING CODE> cd "practical7"
PS C:\Users\AI_LAB\Documents\COMPUTERNETWORK HAMMING CODE\practical7> .\receiver.exe
===== SLIDING WINDOW RECEIVER =====

Receiver is waiting for frames...

-----
Frames received:
-----
Frame No: 0   DATA: c
Frame No: 1   DATA: o
Frame No: 2   DATA: m

All frames received correctly.
Sending ACK:2

-----
Frames received:
-----
Frame No: 3   DATA: p
Frame No: 4   DATA: u
Frame No: 5   DATA: t
```


Problems Output Debug Console **Terminal** Ports

All frames received correctly.
Sending ACK:2

Frames received:

Frame No: 3 DATA: p
Frame No: 4 DATA: u
Frame No: 5 DATA: t

All frames received correctly.
Sending ACK:5

Frames received:

Frame No: 6 DATA: e
Frame No: 7 DATA: r

All frames received correctly.
Sending ACK:7

□